Do AI chatbots improve second language speaking skills?

Yes, **AI chatbots improve speaking skills**, especially fluency, pronunciation, and anxiety reduction.

Consensus Meter N = 16



Yes 63% · Possibly 31% · Mixed 0% · No 6%

FIGURE 1 Scientific consensus on AI chatbots improving second language speaking skills

Empirical research and recent meta-analyses confirm that interacting with conversational AI chatbots systematically improves second language speaking outcomes (Wang et al., 2024; Lyu et al., 2024; Li et al., 2025). Large syntheses report positive medium effect sizes for chatbot interventions over traditional instruction, ranging from $g = 0.484$ to $g = 0.608$ across proficiency levels (Wang et al., 2024; Lyu et al., 2024). These benefits are particularly pronounced for oral fluency, phonological control, and affective factors like communication willingness, although effects on grammatical complexity and listening remain variable (Ding & Yusof, 2025; Oh et al., 2026; Matiienko-Silnytska et al., 2025).

Linguistic Gains Across Speaking Dimensions

Controlled interventions demonstrate that conversational bots foster substantial improvements across several core sub-skills of oral proficiency (Ding & Yusof, 2025; Fathi et al., 2024; Tai & Chen, 2024; Li et al., 2025). In a six-week randomized trial with undergraduate learners preparing for the IELTS exam, chatbot practice led to statistically significant gains in speaking performance compared to traditional instruction (Ding & Yusof, 2025).

Table 3. ANCOVA Results for L2 Speaking Skill . (Ding & Yusof, 2025)

- Speech rate and **uninterrupted run lengths** increase notably during structured chatbot interaction compared to peer dialogues (Matiienko-Silnytska et al., 2025).
- Voice-based systems provide **targeted pronunciation feedback** and phonological refinement through automatic speech recognition (Du & Daniel, 2024; Oh et al., 2026; Li et al., 2025; Lyu et al., 2024).
- Practice facilitates rapid **self-initiated repair sequences** that resolve lexical retrieval stalls without disrupting conversational progressivity (Matiienko-Silnytska et al., 2025).
- Gains concentrate primarily on **fluency and vocabulary**, whereas syntactic accuracy and grammatical complexity often require more explicit, extended interventions (Oh et al., 2026; Li et al., 2025).

Comparison of Chatbot Modalities and Contexts

Interaction Modality	Key System Capabilities	Primary Linguistic Impact	Evidence Caveats
Generative AI bots (Tai &	Open-ended dialogue, dynamic scaffolding,	Promotes **conversational automaticity** and	May produce occasional inaccuracies

Interaction Modality	Key System Capabilities	Primary Linguistic Impact	Evidence Caveats
Chen, 2024; Li et al., 2025; Lyu et al., 2024)	contextual responses (Tai & Chen, 2025; Wang et al., 2024; Lyu et al., 2024)	spontaneous output volume (Tai & Chen, 2025; Taeza, 2025; Lyu et al., 2024)	or lack sociopragmatic depth (Taeza, 2025)
Voice-based systems (Jeon et al., 2023; Oh et al., 2026)	Turn-taking practice, automated speech recognition (Jeon et al., 2023; Oh et al., 2026)	Substantially enhances **speech delivery rate** and pronunciation (Oh et al., 2026; Matiienko-Silnytska et al., 2025; Lyu et al., 2024)	Speech recognition errors can occasionally induce false repairs (Matiienko-Silnytska et al., 2025)
Embodied avatars (Wang et al., 2024; Tai & Chen, 2024)	Anthropomorphic visual presence, synthesized voices (Wang et al., 2024; Tai & Chen, 2025; Lyu et al., 2024)	Boosts **communicative confidence** and reduces speaking inhibition (Wang et al., 2024; Tai & Chen, 2024)	Does not always outperform non-embodied bots on standardized tests (Wang et al., 2024)
Task-guided bots (Ding & Yusof, 2025; Yang et al., 2022; Hsu et al., 2021)	Predefined role-play scenarios, corrective feedback loops (Yang et al., 2022; Hsu et al., 2021; Lee et al., 2024)	Elevates **task completion rates** and focused vocabulary use (Yang et al., 2022; Matiienko-Silnytska et al., 2025)	Formulaic exchanges may limit free-flowing discourse adaptation (Taeza, 2025; Oh et al., 2026)

FIGURE 2 Taxonomy of conversational AI configurations and observed second language learning outcomes

Affective Scaffolding and Implementation Caveats

A central mechanism driving oral improvements is the reduction of foreign language speaking anxiety (Ding & Yusof, 2025; Oh et al., 2026; Lyu et al., 2024). Speaking with an artificial interlocutor provides a non-judgmental, low-stakes environment where learners feel comfortable making mistakes, testing hypotheses, and initiating speech turns (Ding & Yusof, 2025; Tai & Chen, 2024; Oh et al., 2026; Hsu et al., 2021; Matiienko-Silnytska et al., 2025). Interventions combining conversational agents with collaborative structures—such as think–pair–share or paired practice—report significant drops in speaking apprehension paired with elevated foreign language enjoyment and willingness to communicate (Fathi et al., 2024; Tai & Chen, 2024; Taeza, 2025; Wu et al., 2025). Interacting with bots that integrate empathetic or humorous dialogue strategies further supports learner persistence and satisfaction (Zhai & Wibowo, 2022).

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flowchart LR
    A[Voice-Based AI Interaction] --> B[Private, Low-Stakes Practice]
    B --> C[Reduced Speaking Anxiety]
    C --> D[Increased Output Volume & Risk-Taking]
    D --> E[Improved Fluency & Automaticity]
    A --> F[ASR & Immediate Feedback]
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    F --> G[Phonological & Lexical Monitoring]
    G --> E
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Despite these benefits, several pedagogical limitations persist. Current conversational systems often struggle with subtle prosodic feedback, pragmatic nuances, and natural conversational backchanneling (Taeza, 2025; Matiienko-Silnytska et al., 2025). Highly proficient speakers can experience diminished utility when dialogue trees or model responses become repetitive or predictable (Taeza, 2025; Li et al., 2025). Consequently, researchers recommend a blended pedagogical model: utilizing voice-based AI chatbots for scalable rehearsal, lexical automatization, and anxiety reduction, while relying on human instructors and peers to develop complex discourse navigation and socio-pragmatic competence (Taeza, 2025; Matiienko-Silnytska et al., 2025; Li et al., 2025).

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